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Project Report

Fake News Detection and Evaluation with Confusion Matrix

Name	Vamsi Jagannadh Dhenuvakonda
Course	B. Tech in Computer Science and Engineering (Data Science)
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Institute	Mahatma Gandhi Institute of Technology

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Abstract

The objective of this project is to develop a machine learning model capable of classifying news articles as either fake or genuine. The project uses textual data from news articles and applies preprocessing, feature extraction, and classification techniques. Logistic Regression and Decision Tree algorithms were used for classification. The performance of the models was evaluated using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis.

1. Introduction

Fake news has become a major challenge in the digital age. The rapid spread of misinformation through online platforms can influence public opinion and create confusion. Machine learning techniques can help identify fake news automatically by analyzing textual content. This project focuses on building and evaluating a fake news classification system using supervised learning methods.

2. Dataset Description

The dataset consists of two files used for binary classification of news articles:

- fake.csv – Contains fake news articles (labeled as class 1)
- true.csv – Contains genuine news articles (labeled as class 0)

The dataset includes textual news content and was provided as part of the internship project. Both files were merged and shuffled to create a balanced training corpus.

Output of Question 1 (`fake_df.head(3)`)

First three rows of the Fake News dataset loaded into the DataFrame.

title	text	subject	date	
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017

3. Methodology

The project follows a structured nine-step pipeline from raw data ingestion through to model evaluation and hyperparameter tuning:

▸ Step 1: Data Loading

The fake and true news datasets were loaded into separate pandas DataFrames using Python's pandas library.

▸ Step 2: Data Labeling

A new column named "class" was added to each DataFrame to assign binary labels:

- Fake news = 1
- True news = 0

▸ Step 3: Data Merging and Shuffling

Both datasets were combined into a single DataFrame and randomly shuffled to eliminate ordering bias before training.

▸ Step 4: Data Cleaning

Unnecessary columns — title, subject, and date — were removed, retaining only the article text and its class label.

▸ Step 5: Text Preprocessing

The text was converted to lowercase. Special characters, URLs, punctuation, and unwanted symbols were removed using regular expressions to produce clean text inputs.

▸ Step 6: Train-Test Split

The dataset was divided into training and testing sets (80/20 split) to allow unbiased model evaluation on unseen data.

▸ Step 7: Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) Vectorization was applied to convert the cleaned text into numerical feature vectors suitable for machine learning models.

▸ Step 8: Model Training

The following supervised machine learning algorithms were trained on the TF-IDF feature vectors:

- Logistic Regression – a probabilistic linear classifier well-suited for binary text classification
- Decision Tree Classifier – a non-linear model that partitions the feature space through recursive binary splits

▸ Step 9: Hyperparameter Tuning

GridSearchCV was used to systematically search for the best hyperparameters for the Logistic Regression model, optimizing cross-validated accuracy over a predefined parameter grid.

4. Results

Figure 2: Classification report generated by the Logistic Regression model showing precision, recall, F1-score, and support values.

	precision	recall	f1-score	support
0	0.98	1.00	0.99	5373
1	0.99	0.97	0.98	2881
accuracy			0.99	8254
macro avg	0.99	0.98	0.99	8254
weighted avg	0.99	0.99	0.99	8254

Decision Tree classifier accuracy obtained on the test dataset:

0.9970923188756966 (approx. 99.71%)

GridSearchCV output:

{'C': 10, 'solver': 'liblinear'}

Both models demonstrated outstanding classification performance on the held-out test set. The table below summarizes key evaluation metrics:

Metric	Logistic Regression	Decision Tree	Remarks
Accuracy	~99%	99.71%	Excellent
Precision	High	High	Low false positives
Recall	High	High	Low false negatives
F1-Score	High	High	Balanced metric
Best Param (C)	10	N/A	GridSearchCV tuned
Best Solver	liblinear	N/A	Optimal for text

The Logistic Regression model achieved approximately 99% classification accuracy with high precision, recall, and F1-score values across both classes. The best parameters identified by GridSearchCV were C = 10 and solver = liblinear.

The Decision Tree Classifier achieved an accuracy of 99.71%, slightly outperforming Logistic Regression on raw accuracy while maintaining equally high precision and recall values. Both models produced well-balanced confusion matrices with minimal misclassification.

5. Conclusion

The project successfully developed a machine learning-based fake news detection system. Both Logistic Regression and Decision Tree classifiers achieved excellent performance with accuracies exceeding 99%. The use of TF-IDF vectorization combined with supervised learning algorithms proved highly effective for text classification.

The project demonstrates how machine learning can be applied to combat the growing problem of misinformation by automatically identifying fake news articles at scale. Future work could explore deep learning models such as LSTM or BERT for potentially even higher accuracy and better generalization across diverse news domains.

References

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